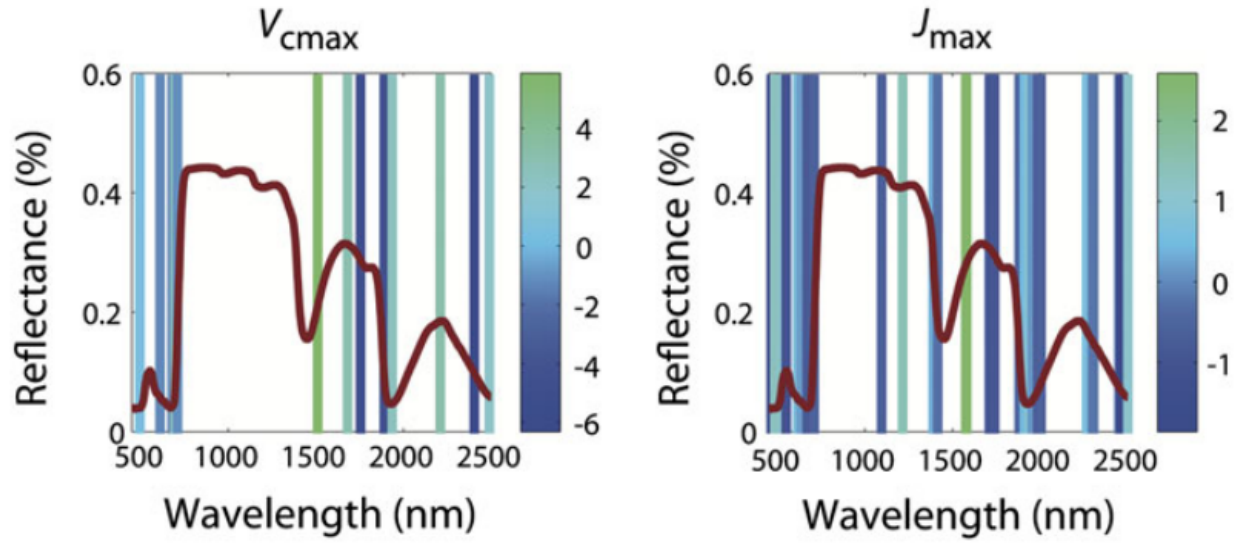


RESEARCH PAPER

Leaf optical properties reflect variation in photosynthetic metabolism and its sensitivity to temperature

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- [10.1093/jxb/err294](http://dx.doi.org/10.1093/jxb/err294)
- PLSR with full range spectrometers on leaves of trembling aspen and eastern cottonwood trees grown across an array of glasshouse temperature regimes.
- The wavelengths selected were correlated with known absorption features related to leaf water content, nitrogen concentration, internal structure, and/or photosynthetic enzymes.
- 13 wavelengths identified for $V_{\max}$ and 44 wavelengths for $J_{\max}$ (Figure 1).
- “Use of full-spectrum data can often result in PLSR models with spectral loadings that do not contribute significantly to the prediction (i.e. are near 0) and may negatively affect results.”



V_{cmax} ($\mu\text{mol m}^{-2} \text{s}^{-1}$)

Parameter	Raw PLSR Coefficient
Intercept	-105.105
<i>Wavelength</i>	
495	2512.749
610	-3804.889
680	-9121.729
695	8114.260
710	-1281.079
1510	4284.383
1680	6566.049
1755	-9975.176
1890	-5262.651
1935	8315.752
2210	7361.025
2405	-14208.388
2490	7473.886
Jackknife statistics:	
Median $R^2 = 0.78$	
Median RMSE = 16.2	

J_{max} ($\mu\text{mol m}^{-2} \text{s}^{-1}$)

Parameter	Raw PLSR Coefficient
Intercept	-20.650
<i>Wavelength</i>	
465	-274.189
485	3984.520
535	5125.756
550	-5721.735
615	4191.832
635	1109.040
670	-10882.564
710	-428.916
1095	-4967.966
1215	6337.985
1390	5793.293
1415	-5534.300
1575	4536.632
1580	4826.890
1710	-1167.044
1715	-1833.546
1720	-1927.169
1725	-2214.091
1730	-2548.040
1740	-1763.486
1880	-2488.854
1885	-1480.018
1890	-239.556
1895	837.299
1910	3433.131
1935	2952.828
1940	2893.471
1955	2376.691
1985	-271.257
1990	-1181.458
1995	-733.476
2000	-2350.489
2005	-1752.527
2265	1241.956
2270	581.440
2275	1592.081
2280	2434.431
2290	296.427
2295	316.057
2300	-2038.929
2305	-969.998
2450	2103.843
2460	-6022.569
2500	2491.480
Jackknife statistics:	
Median $R^2 = 0.77$	
Median RMSE = 20.1	

Figure 1: Selected wavelengths for V_{cmax} and J_{max} PLSR models (top), and their raw PLSR coefficients (bottom). The coefficient tables are from the SI of the paper.